

A Workshop for Understanding and Using Instrument-free Methods

When Gaussian Copulas Work, When They Crack and a Novel Approach for Taking
Up the Pieces

Special Session / Workshop @ EMAC 26, Bath, UK

03 June 2026

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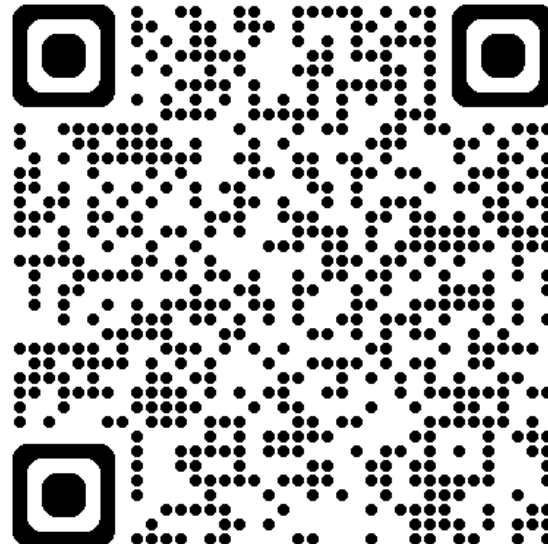
Agenda and Materials

Our plan for today:

- 1) Opening: *5min*
- 2) Introduction to the Gaussian Copula method: *15 min*
- 3) Using Gaussian Copula methods in R: *20 min*
- 4) Break: *5 min*
- 5) Revisiting the assumptions in a DAG; introducing new method: *10 min*
- 6) *Introducing a new method (ICA) 10 min.*
- 7) Using the new method in R: *10 min*
- 8) *What to do as author / reviewer? 5 min*
- 9) Discussion, Q&A, buffer: *10 min*

Materials:

<https://github.com/HashtagHaschka/Workshop>



Other stuff you might want or need:

- Papers on Copula:
 - Haschka, R. E. (2022). Handling endogenous regressors using copulas: a generalization to linear panel models with fixed effects and correlated regressors. *Journal of Marketing Research*, 59(4), 860-881.
 - Haschka, R. E. (2025). Robustness of copula-correction models in causal analysis: Exploiting between-regressor correlation. *IMA Journal of Management Mathematics*, 36(1), 161-180.

Recent paper when Copula cracks:

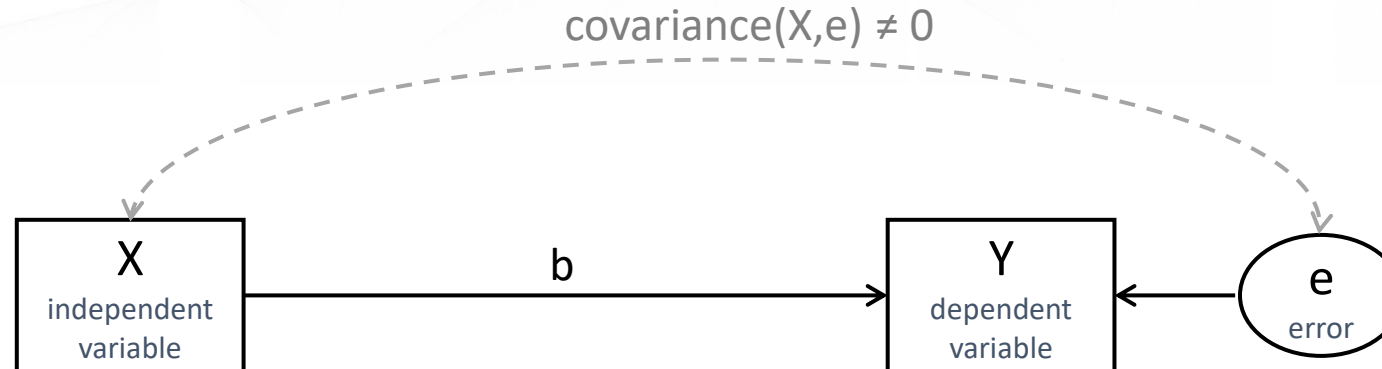
Dost, Florian and Haschka, Rouven E., The Gaussian Copula Control Function Method Does Not Help Against Traditional Omitted Variable Bias (June 07, 2025). Available at SSRN: <http://dx.doi.org/10.2139/ssrn.5285127>

New instrument-free method for omitted variables bias

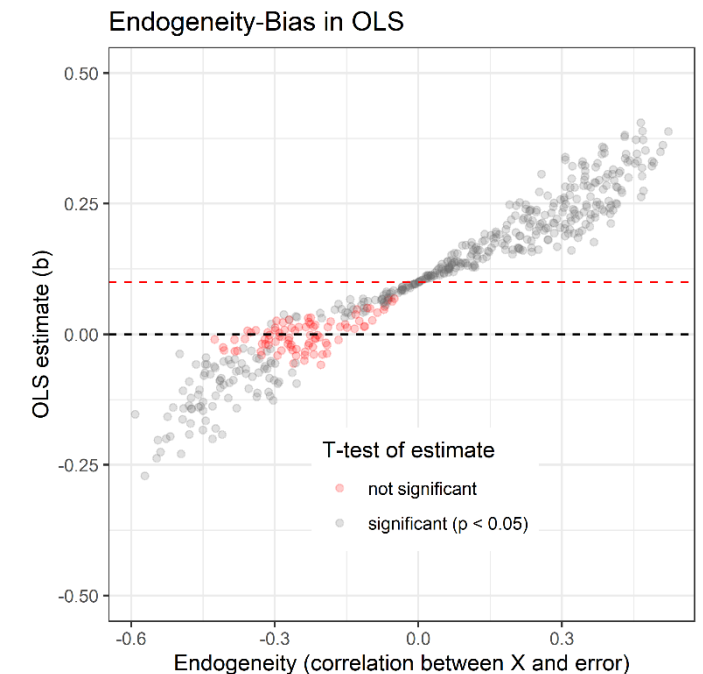
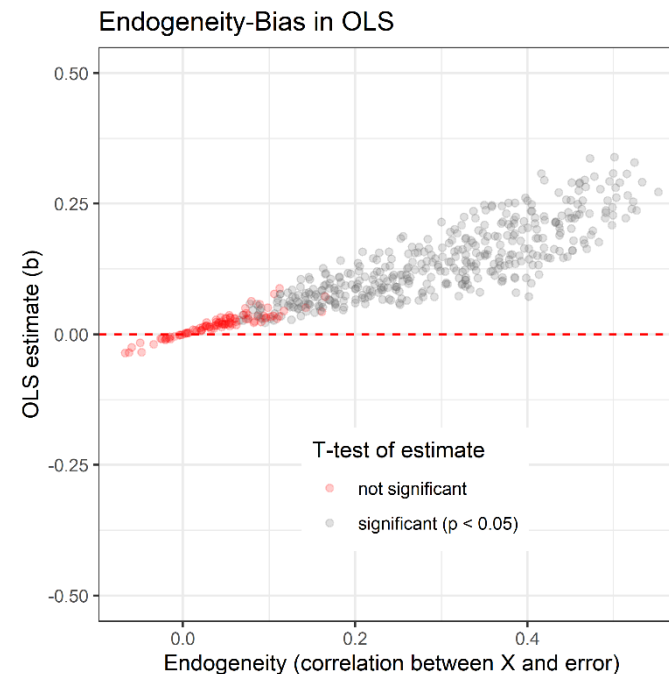
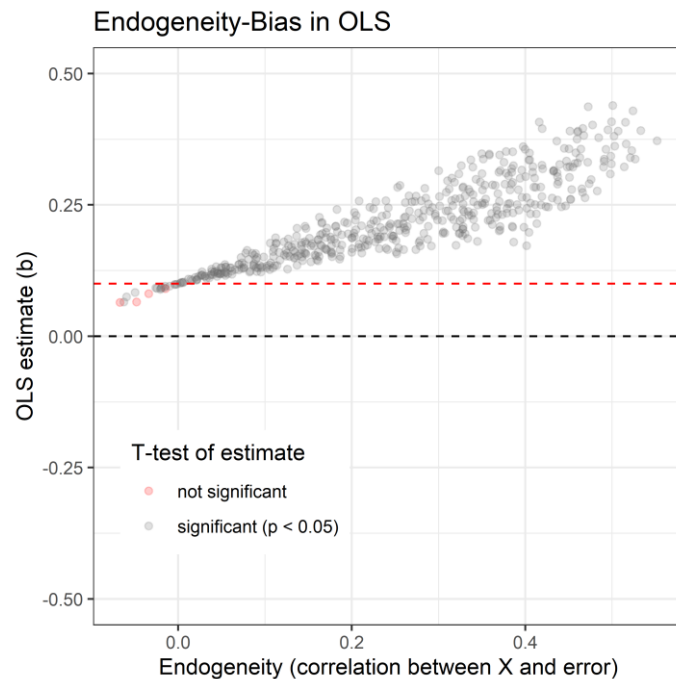
Haschka, Rouven E. and Dost, Florian, ICA at the Cocktail Party: Casting Instrument-free Omitted Variable Bias Correction as a Blind Source Separation Problem (July 22, 2025). Available at SSRN: <http://dx.doi.org/10.2139/ssrn.5361801>

What is the problem with endogeneity?

What is endogeneity?

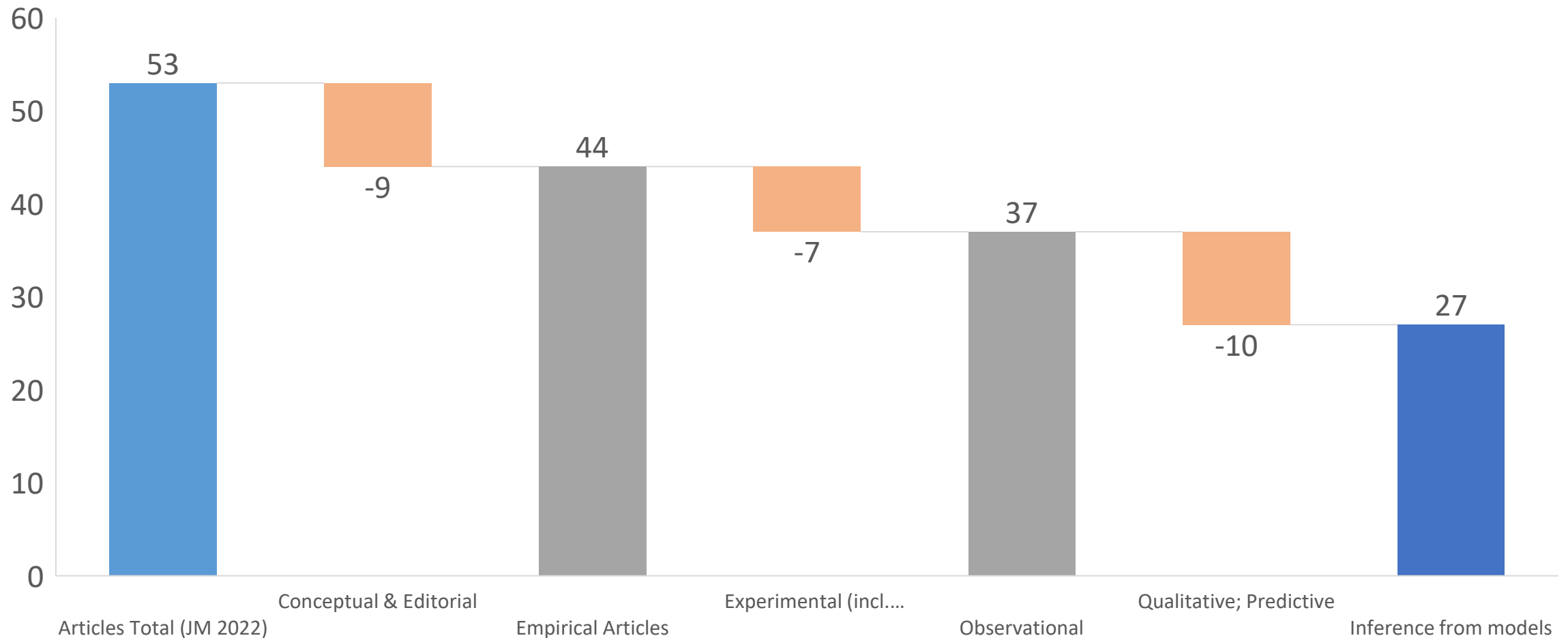


Why does it matter?



Who cares about endogeneity?

Articles in the Journal of Marketing, in 2022, by type of included studies



Who else cares about endogeneity?

A selection of recent articles in Marketing and Management journals about endogeneity:

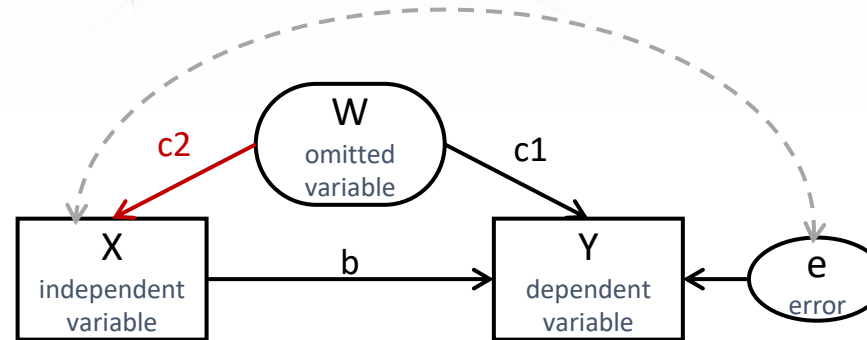
- Becker, J. M., Proksch, D., & Ringle, C. M. (2022). Revisiting Gaussian copulas to handle endogenous regressors. *Journal of the Academy of Marketing Science*, 50(1), 46-66.
- Busenbark, J. R., Yoon, H., Gamache, D. L., & Withers, M. C. (2022). Omitted variable bias: Examining management research with the impact threshold of a confounding variable (ITCV). *Journal of Management*, 48(1), 17-48.
- Eckert, C., & Hohberger, J. (2022). Addressing endogeneity without instrumental variables: An evaluation of the gaussian copula approach for management research. *Journal of Management*, 01492063221085913.
- Haschka, R. E. (2022). Handling endogenous regressors using copulas: A generalization to linear panel models with fixed effects and correlated regressors. *Journal of Marketing Research*, 59(4), 860-881.
- Hill, A. D., Johnson, S. G., Greco, L. M., O'Boyle, E. H., & Walter, S. L. (2021). Endogeneity: A review and agenda for the methodology-practice divide affecting micro and macro research. *Journal of Management*, 47(1), 105-143.
- Li, J., Ding, H., Hu, Y., & Wan, G. (2021). Dealing with dynamic endogeneity in international business research. *Journal of International Business Studies*, 52, 339-362.
- Lu, G., Ding, X. D., Peng, D. X., & Chuang, H. H. C. (2018). Addressing endogeneity in operations management research: Recent developments, common problems, and directions for future research. *Journal of Operations Management*, 64, 53-64.
- Qian, Y., & Xie, H. (2023). Correcting Regressor-Endogeneity Bias via Instrument-Free Joint Estimation Using Semiparametric Odds Ratio Models. *Journal of Marketing Research*.
- Rutz, O. J., & Watson, G. F. (2019). Endogeneity and marketing strategy research: An overview. *Journal of the Academy of Marketing Science*, 47, 479-498.
- Sajons, G. B. (2020). Estimating the causal effect of measured endogenous variables: A tutorial on experimentally randomized instrumental variables. *The Leadership Quarterly*, 31(5), 101348.
- Ullah, S., Zaefarian, G., & Ullah, F. (2021). How to use instrumental variables in addressing endogeneity? A step-by-step procedure for non-specialists. *Industrial Marketing Management*, 96, A1-A6.
- Wolfolds, S. E., & Siegel, J. (2019). Misaccounting for endogeneity: The peril of relying on the Heckman two-step method without a valid instrument. *Strategic Management Journal*, 40(3), 432-462.

The problem is everywhere: sources of endogeneity

- Omitted variable bias:

Examples:

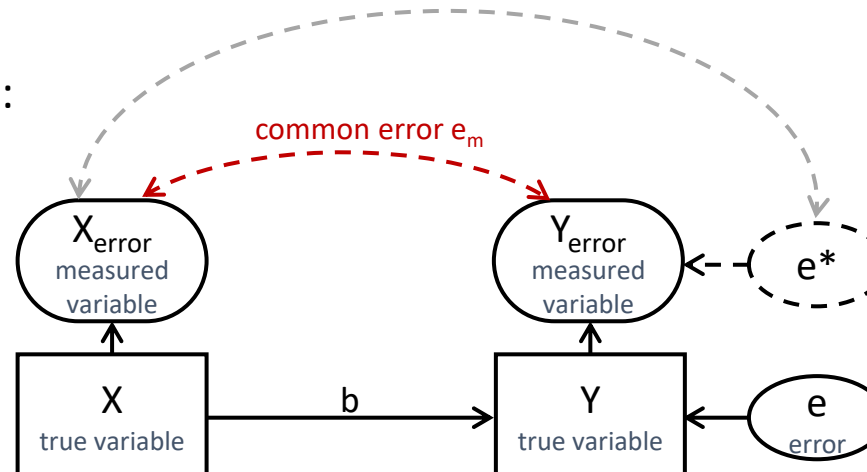
- Ability affects selection into training, both ability and training affect performance.
- Events (superbowl) increase ad effectiveness; managers plan accordingly; observed ad effects are higher.
- Events (conference) increase hotel demand; managers increase price; price and demand correlate positively.



- Measurement error (CMV or attenuation):

Examples:

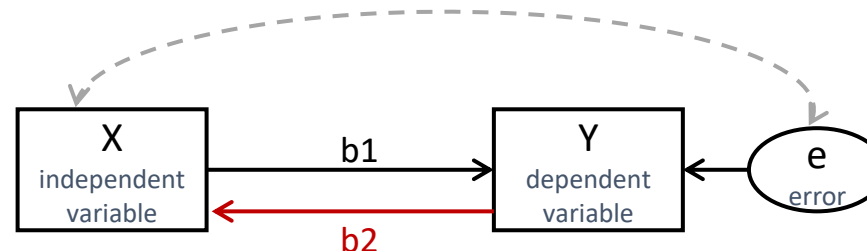
- Abilities, job satisfaction and perceived job performance are measured together in a short survey
- Footfall tracking in a store is imprecise on busy days. (a bias, but not necessarily endogeneity)



- Simultaneity:

Examples:

- Job performance and job satisfaction.
- Marketing budgets and firm performance.
- Social media buzz and brand equity.
- Demand and supply set prices.



The Gaussian Copula endogeneity correction is widely used in marketing

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Addressing Endogeneity Without Instrumental Variables: An Evaluation of the Gaussian Copula Approach for Management Research

Christine Eckert
University of Technology Sydney
Jan Hohberger
University Ramon Llull

Article

Addressing Endogeneity Using a Two-Stage Copula Generated Regressor Approach

Fan Yang, Yi Qian, and Hui Xie

Journal of the Academy of Marketing Science (2025) 53:279–299
https://doi.org/10.1007/s11747-024-01055-4

METHODOLOGICAL PAPER

Dealing with regression models' endogeneity by means of an adjusted estimator for the Gaussian copula approach

Benjamin D. Liengard, Jan-Michael Becker, Mikkel Bennedsen, Phillip Heiler, Luke N. Taylor, Christian M. Ringle

Handling Endogenous Regressors by Joint Estimation Using Copulas

Sungho Park
W. P. Carey School of Business, Arizona State University, Tempe, Arizona 85287, spark104@asu.edu
Sachin Gupta
Samuel Curtis Johnson Graduate School of Management, Cornell University, Ithaca, New York 14853, sg248@cornell.edu

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Journal of the Academy of Marketing Science (2022) 50:46–66
https://doi.org/10.1007/s11747-021-00805-y

METHODOLOGICAL PAPER

Revisiting Gaussian copulas to handle endogenous regressors

Jan-Michael Becker, Dorian Proksch, Christian M. Ringle

Received: 23 September 2020 / Accepted: 29 July 2021 / Published online: 11 October 2021
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Journal of Marketing

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Impact Factor: 10.4 / 5-Year Impact Factor: 17.9

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EXPRESS: A Practical Guide to Endogeneity Correction Using Copulas

Yi Qian, Anthony Koschmann, and Hui Xie View all authors and affiliations

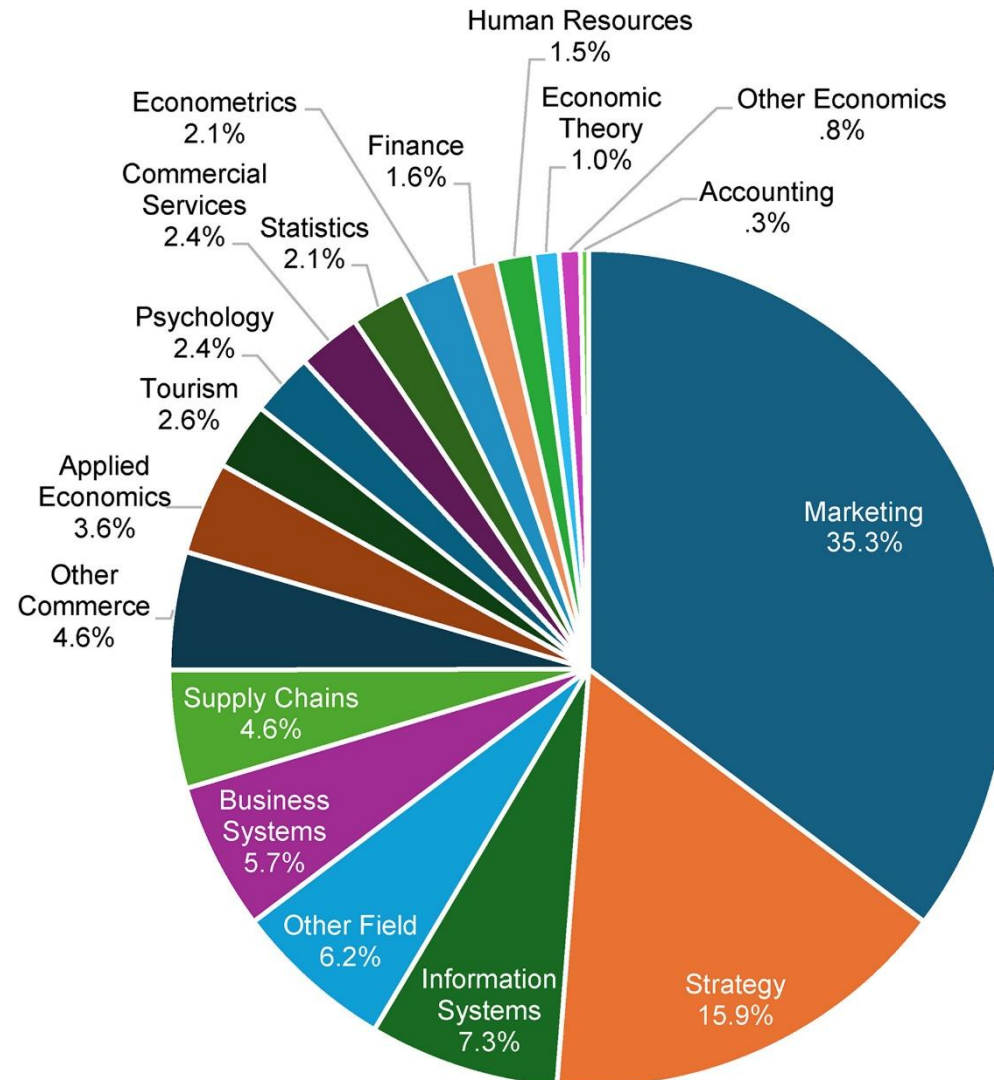
Accepted Manuscripts | https://doi.org/10.1177/00222429251410844

GCCF use by discipline (from Qian, Koschmann & Xie 2025)

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A: Publications (n = 615) Using Copula Correction by Disciplines



What researchers use GCCF for? Omitted variable bias...

Literature using GCCF:

- **82** Articles using GCCF for identification
- Published before 01/06/2024.
- Journals searched: *Journal of Marketing, Journal of Marketing Research, Marketing Science, Journal of Consumer Research, Journal of the Academy of Marketing Science, Journal of Retailing, International Journal of Research in Marketing, Journal of Consumer Psychology, International Journal of Information Management, Academy of Management Perspectives, Management Science, Journal of Interactive Marketing, Quantitative Marketing and Economics.*

Nature of endogeneity problem:

- Of 82 articles, 50 detail the assumed data-generating process.
- Of those, 46 articles **(92%) assume omitted variables bias**

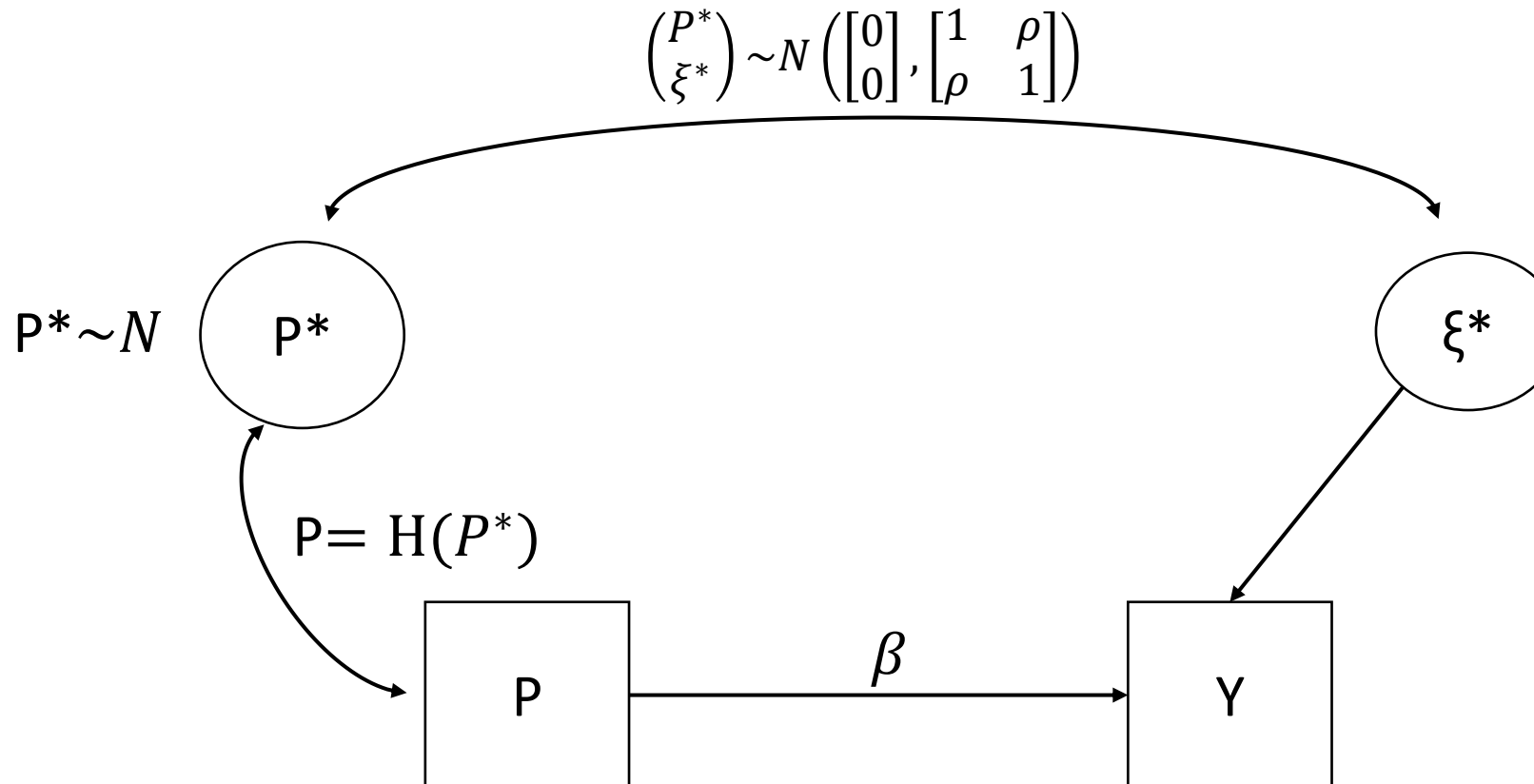
Explicit assumptions for Gaussian Copula Control Function (GCCCF) methods

1st assumption: Joint dependence is bivariate-normal, with correlation ρ , such that a Gaussian copula captures the dependence

$$\begin{pmatrix} P^* \\ \xi^* \end{pmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right)$$

2nd assumption:
Error term ξ is normally distributed

$$\xi^* \sim N$$



3rd assumption:

Regressor P is sufficiently non-normally distributed $P \not\sim N$

$$Y = \alpha + \beta P + \xi$$

$$\xi = \xi^* \sigma$$

Let the Copula Expert continue...

Explicit *and implicit* assumptions for Gaussian Copula Control Function (GCCF) methods

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1st assumption: Joint dependence is bivariate-normal, with correlation ρ , such that a Gaussian copula captures the dependence

$$\begin{pmatrix} P^* \\ \xi^* \end{pmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right)$$

2nd assumption:
Error term
 ξ is normally
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$$\xi^* \sim N$$

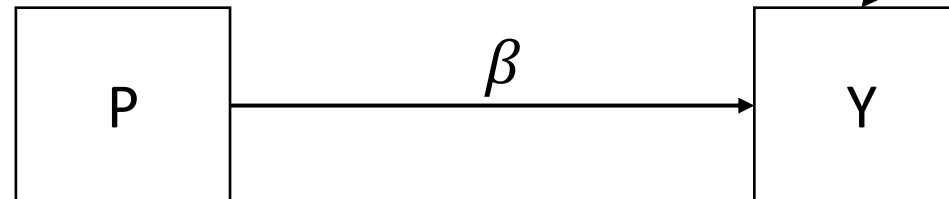
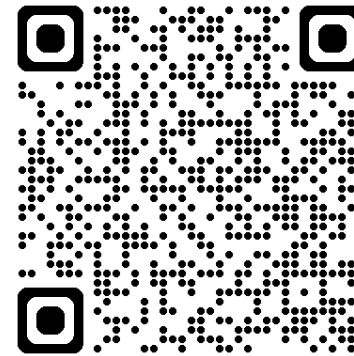
$$P^* \sim N$$

4th, an implicit assumption:
 $H()$ is a bijective and non-linear one-to-one mapping between normally distributed P^* and non-normally distributed P

$$P = H(P^*)$$

3rd assumption:

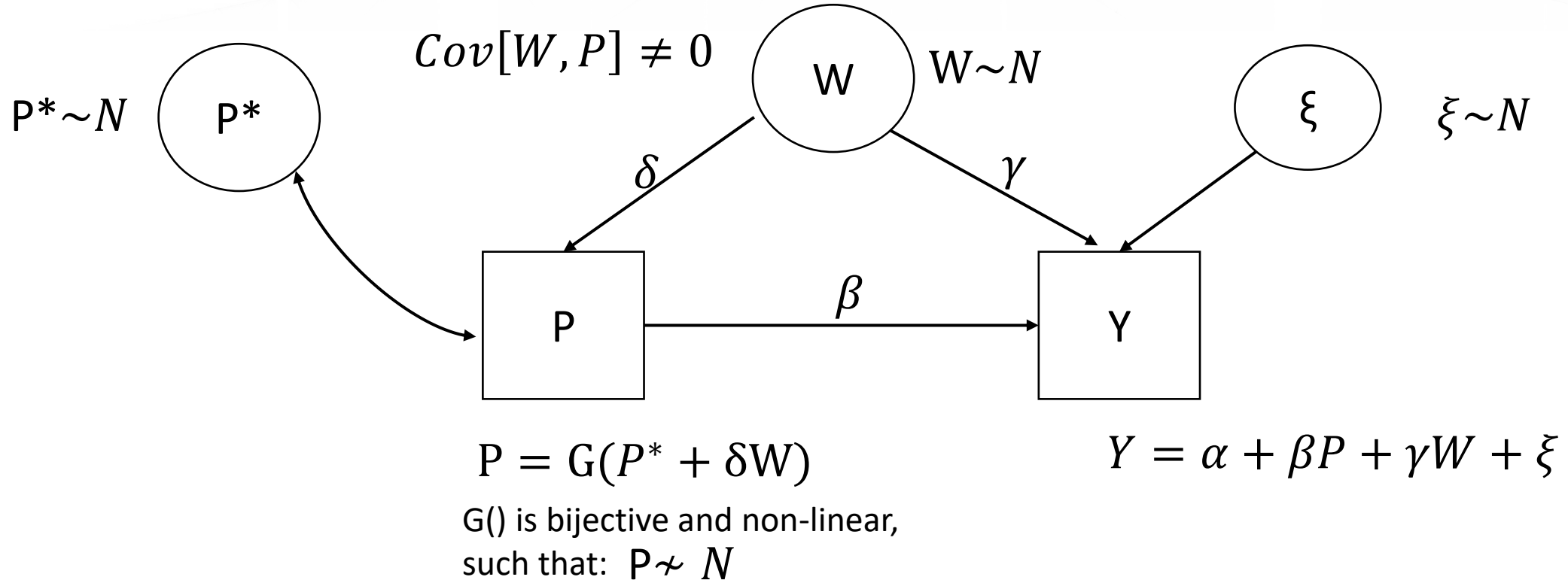
Regressor P is sufficiently non-normally distributed $P \not\sim N$



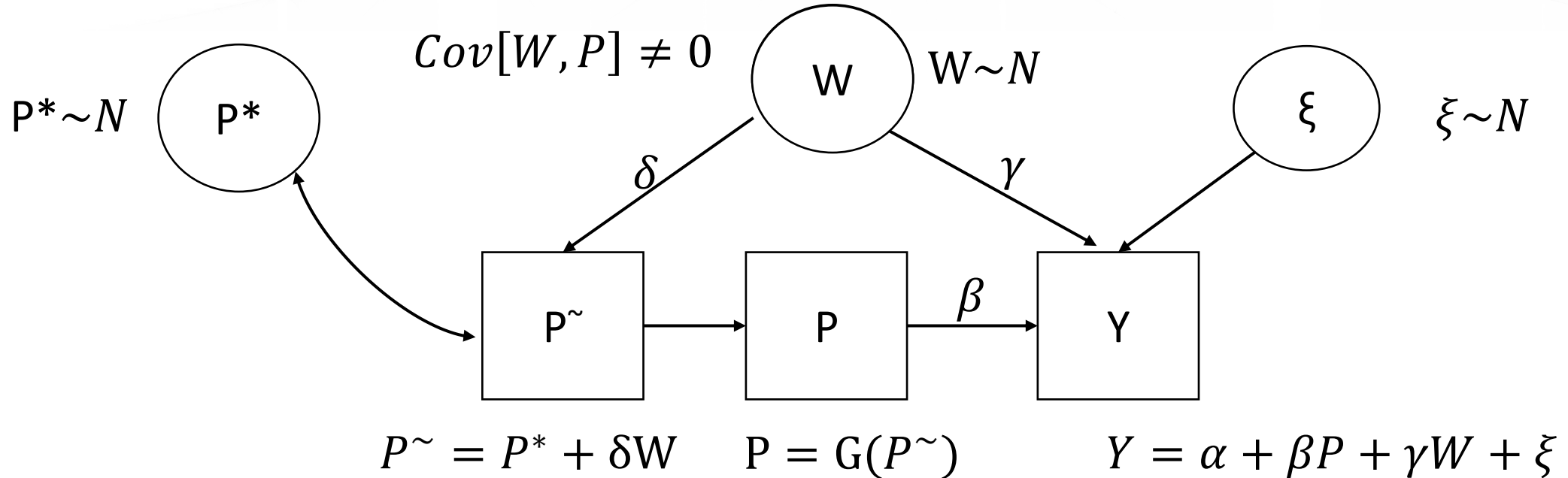
$$Y = \alpha + \beta P + \xi$$

$$\xi = \xi^* \sigma$$

Omitted variable DGP with additive process and nonlinear transform



Equivalent notation with intermediate step:

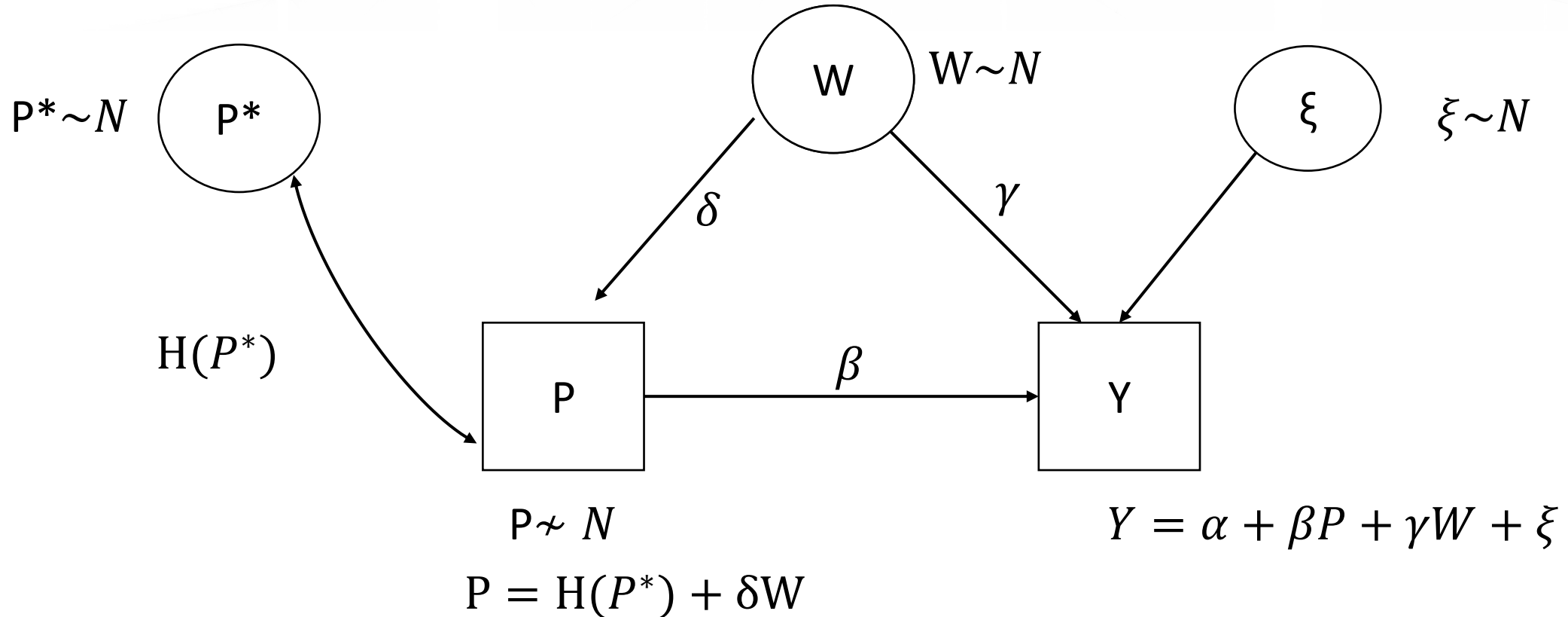


Necessary!
 $G()$ is bijective and nonlinear

Traditional omitted variable DGP with non-Gaussian exogenous component

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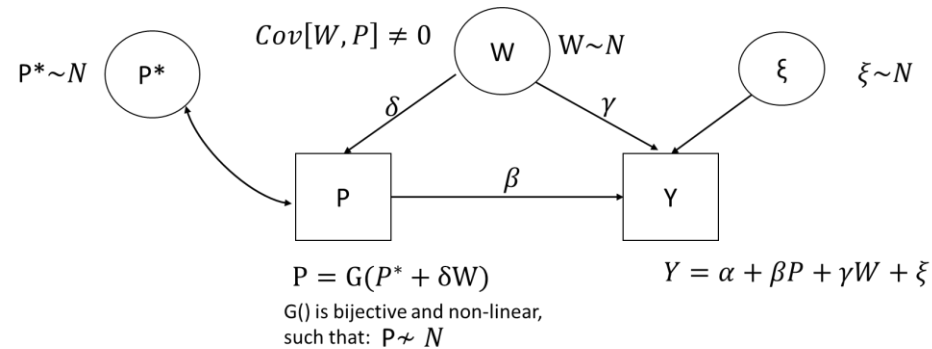
Prove that GCCF estimation
creates a bias:

$$\text{Bias} = \gamma \cdot \frac{\delta \sigma_W^2 - \text{Cov} \left(\mathbb{E}[P_t | \tilde{P}^*], \mathbb{E}[W_t | \tilde{P}^*] \right)}{\mathbb{E}[\text{Var}(P_t | \tilde{P}^*)]}$$

Two simulation studies for illustration

Scenario A:

Omitted variable with bijective transform

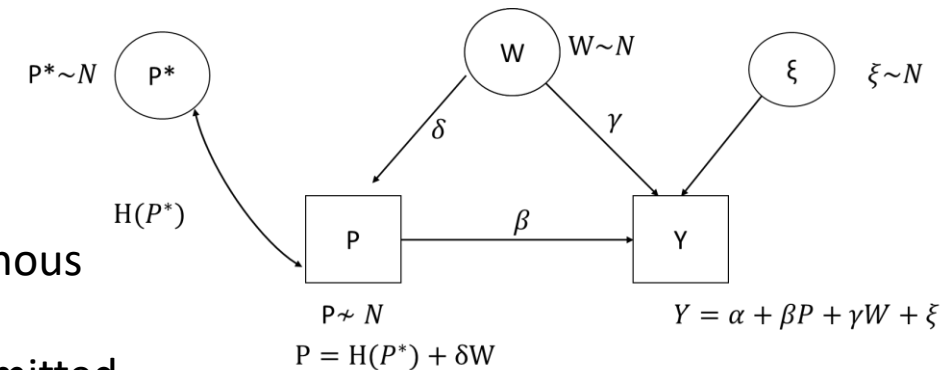


$P_t = G(P_t^* + \delta W_t)$ is the
observed variable

- $Y_t = \alpha + \beta P_t + \gamma W_t + \xi_t$,
- ξ_t is structural error with $E[\xi_t | P_t, W_t] = 0$, and $\xi_t \sim N(0, 1)$,
- $\alpha = 0, \beta = 1, \gamma = 1$,
- $n = 1000$ sample size, $r = 500$ repetitions
- $P_t^* \sim N(0, 1)$ is exogenous variance,
- $W_t \sim N(0, 1)$ is the omitted variable,
- δ is randomly drawn from $[0, 0.4]$ to induce endogeneity.
- $G() = H() = \Phi^{-1}()$ the canonical nonlinear transform to uniform.
- **Bias** = $\beta_{\text{true}} - \beta_{\text{OLS/GCCF}}$

Scenario B:

Traditional omitted variable



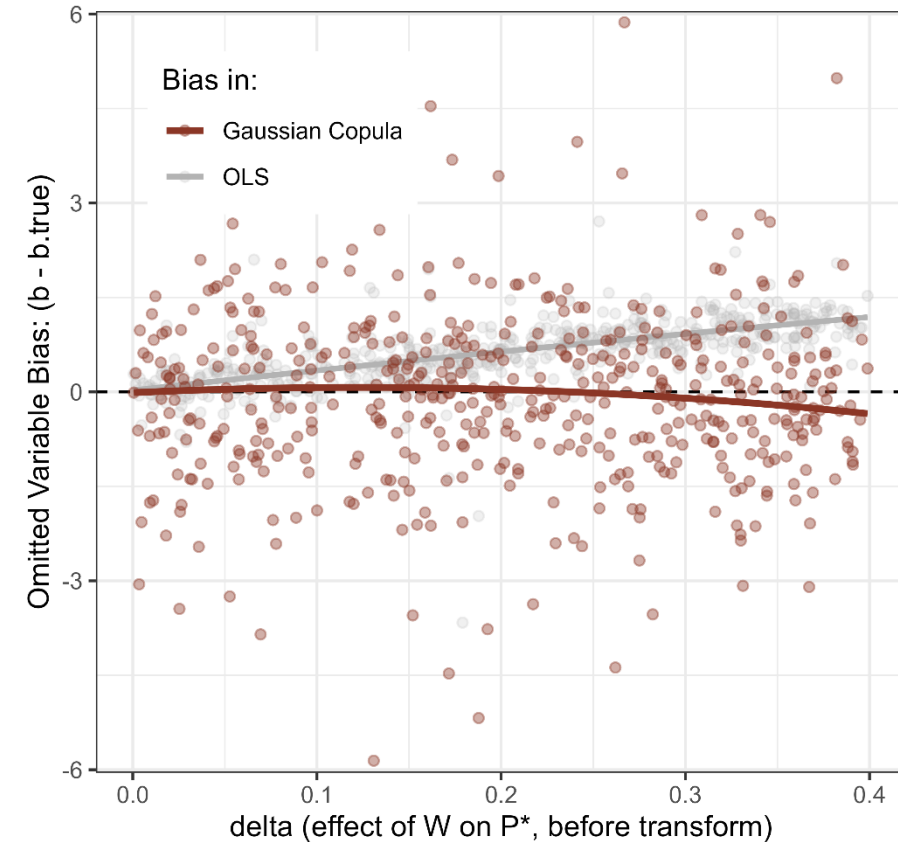
$P_t = H(P_t^*) + \delta W_t$ is the
observed variable

Simulation studies results:

Scenario A:

Omitted variable with bijective transform

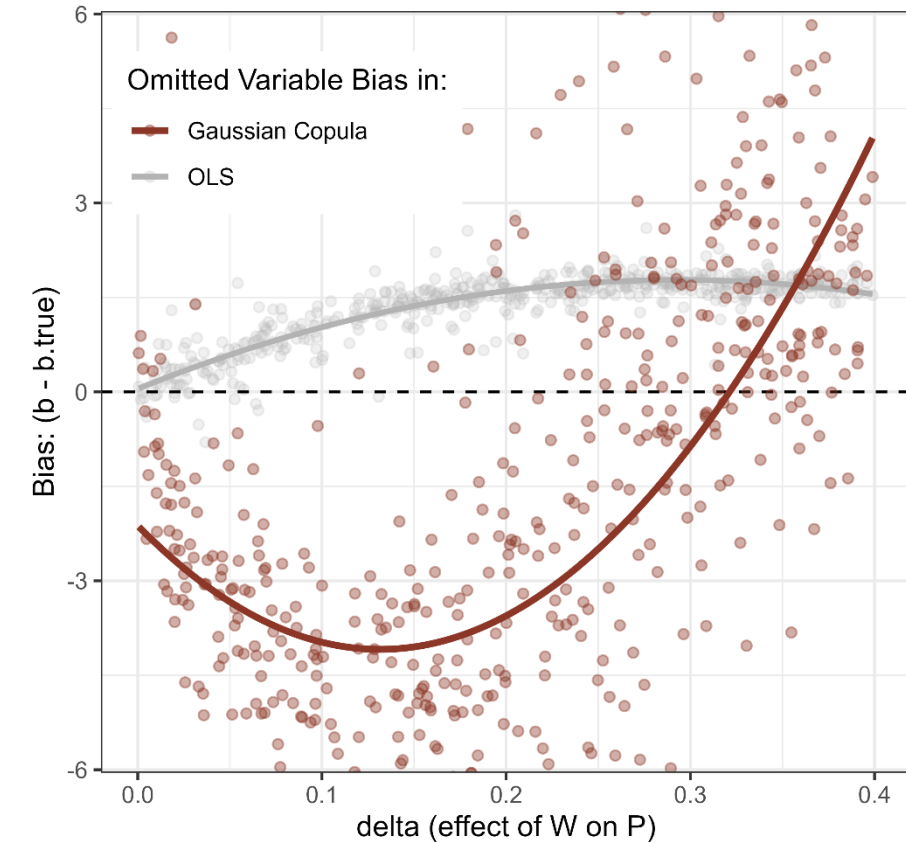
ScenarioA (transform: $G(P^* + \delta W)$)



Scenario B:

Traditional omitted variable

ScenarioB (traditional additive omitted variable)



Recommendations for GCCF use:

TLDR: Explicitly answer: Why is P **endogenous**? **AND:** Why is P **non-normal**?

- **1. Diagnose the DGP first, estimate second.** Begin every empirical project by writing down a verbal or formal data-generating process. If the story is “unobserved trait W adds to both P and the error,” copula correction is contraindicated.
- **2. Treat marginal non-normality as necessary, never sufficient.** Confirming that P is non-normal merely clears only one of several hurdles; it does not establish that a monotone transform of the endogenous component drives P .
- **3. Document the monotone-transform rationale.** When using any copula method, spell out why the underlying mechanism should be nonlinear and strictly monotone (e.g., saturation, diminishing returns, ranking processes).
- **4. Document the order of additions and transforms in the DGP.** Importantly, the transform needs to happen after the omitted variable additively affected the exogenous variance in P .
- **5. Triangulate with IVs when possible.** If a credible instrument is available, compare IV and GCCF estimates. Divergence is an immediate red flag that the monotone-transform assumption may be violated.

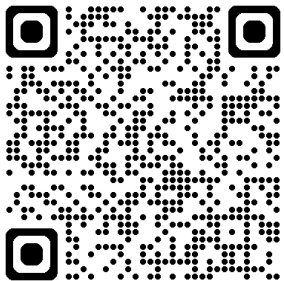
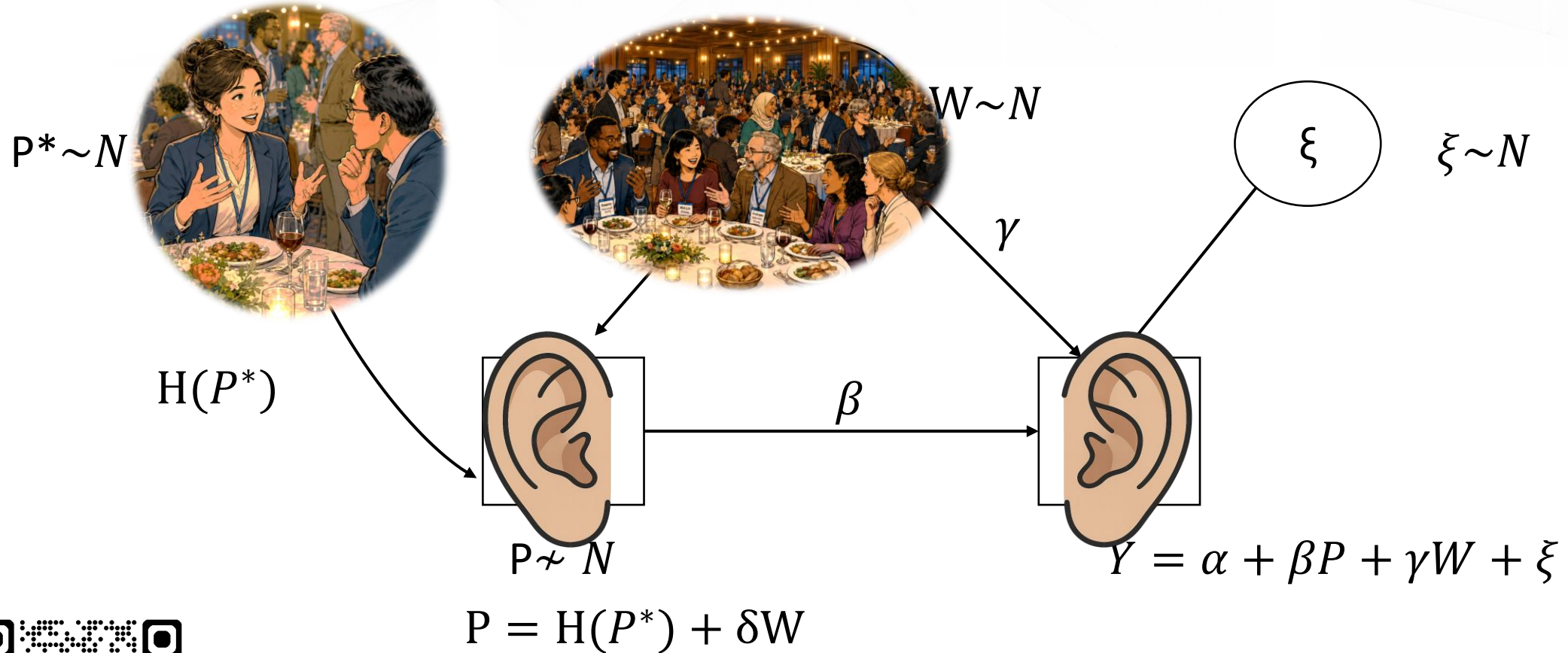
How to spot „additive DGP language“

Synthetic Formulation of Rationales	The Structural Mismatch with GCCF	Relative Frequency
Extensive Covariates <i>“To minimize omitted variable bias, we include an extensive set of control variables.”</i>	Additive Controls vs. Non-linear Residuals: Modeling confounders as additive factors while assuming the residual error requires a non-linear correction creates a bifurcated, inconsistent DGP.	Very High
Absorbing Heterogeneity <i>“To control for unobserved heterogeneity, we include a set of firm, industry, and year fixed effects.”</i>	Additive Shifts vs. Non-linear Copulas: Fixed effects treat unobservables as additive linear intercept shifts. Using GCCF for the remaining error assumes a conflicting non-linear structure for the same DGP.	Very High
Mitigating Endogeneity <i>“The inclusion of fixed effects resolves potential endogeneity issues arising from time-invariant omitted variables.”</i>	Separability vs. Joint Dependence: If endogeneity is “resolved” by additive subtraction, the non-linear dependence required for GCCF identification is theoretically absent.	High
Dimensionality Reduction <i>“We apply PCA to our set of controls to reduce dimensionality.”</i>	Linear Redundancy vs. Non-linear Dependence: PCA assumes that entering variables are linearly related. This contradicts the GCCF requirement that unobservables must have a non-linear dependence with observables.	Moderate
Transforming Data <i>“We employ first-differencing to eliminate persistent unobserved factors from the model residual.”</i>	Linear Cancellation vs. Non-linear Persistence: Linear transformations to cancel unobservables assume a purely additive structure, whereas GCCF requires a non-linear relationship.	Low

Reframe the omitted variable as “the cocktail party problem” from signal processing

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- Implementation of an estimate for W as control function (R code)
- Extensive simulation evidence for robustness in wide array of DGPs
- canonical empirical cases

Using Independent Component Analysis (ICA) to disentangle exogeneous from omitted signal

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- ICA:

- Blind source separation technique
- Based on whitening and rotation
- Several variants (e.g., JADE)

- ICA assumptions:

- As many (or more) measured signals as independent latent sources
- Each signal is a linear combination of sources
- Ideally, all sources are uncorrelated and non-normal
- At most one source can be normal

- Additional assumptions for our use case:

- The exogeneous signal is non normal; the omitted signal (source of endogeneity) is normal
→ ICA is blind, we need the assumption to decide which component is exogenous

Assumption 1. Endogenous regressor can be additively decomposed into an exogenous non-normal part and an endogenous normal part.

Assumption 2. The structural error is either (i) normally distributed, or (ii) additively decomposable into an endogenous normal part and an exogenous part.

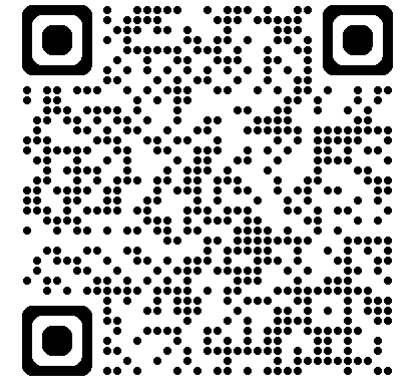
Assumption 3. The endogenous normal part enters the regressor and the structural error linearly.

Assumption 4. Full rank of all regressors and conditional exogeneity of exogenous regressors.

Table 3: Identifying assumptions in the ICA endogeneity correction.

Code and paper (with identification proof) b-tu

```
15 # Step 3: Apply Independent Component Analysis (ICA)
16 data_matrix <- cbind(Y, P)
17 ica_result <- ica(X = data_matrix, nc = 2, method = "jade")
18 # Step 4: Find the ICA component that looks most normally
    distributed
19 hist(ica_result$S[, 1])
20 hist(ica_result$S[, 2])
21 # Step 5: Estimate the regression model, controlling for the normal
    component
22 model <- lm(Y ~ P + ica_result$S[, 1])
23 # View the results
24 summary(model)
```

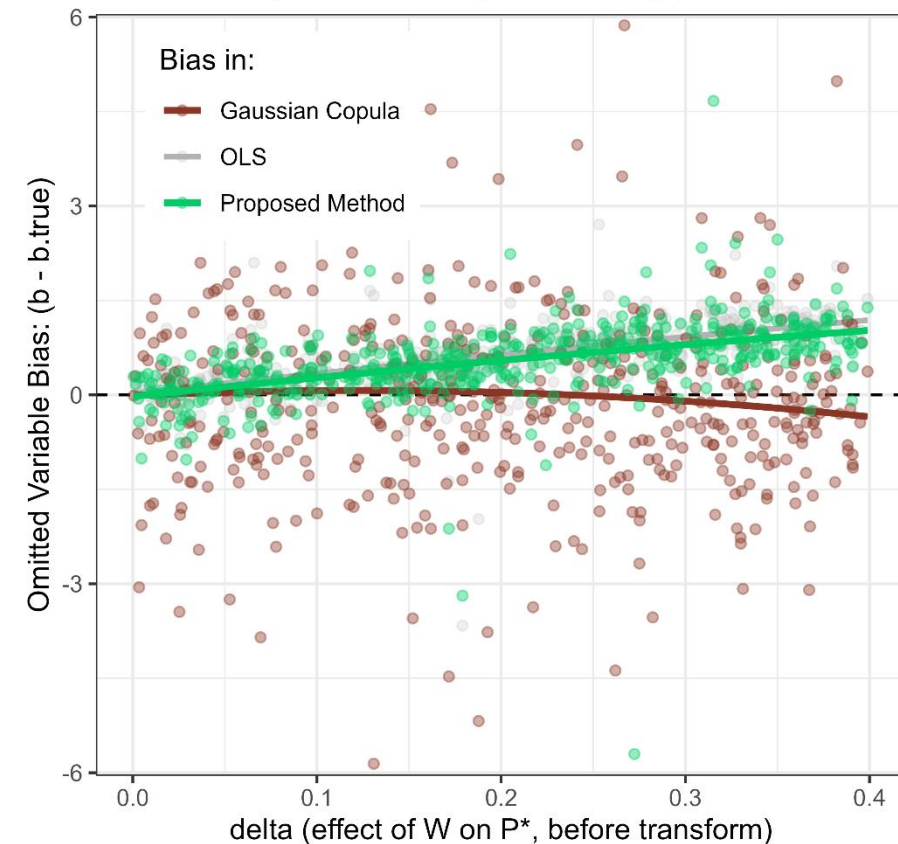


Revisit simulation studies:

Scenario A:

Omitted variable with bijective transform

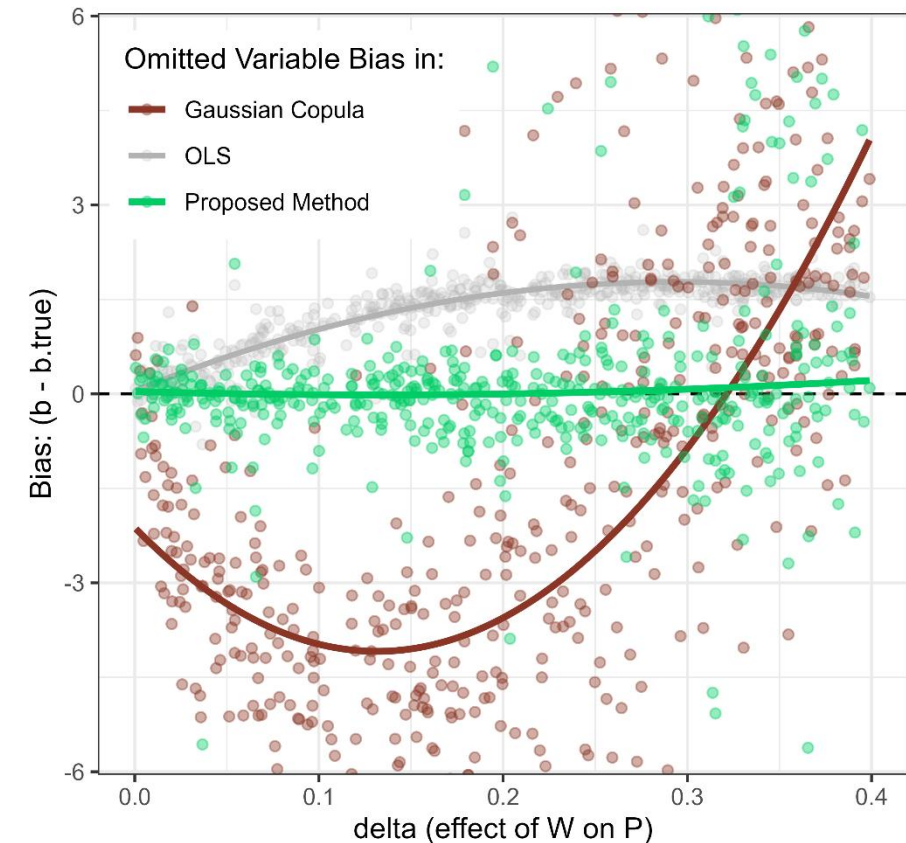
ScenarioA (transform: $G(P^* + \delta W)$)



Scenario B:

Traditional omitted variable

ScenarioB (traditional additive omitted variable)



The author's guide to do ICA

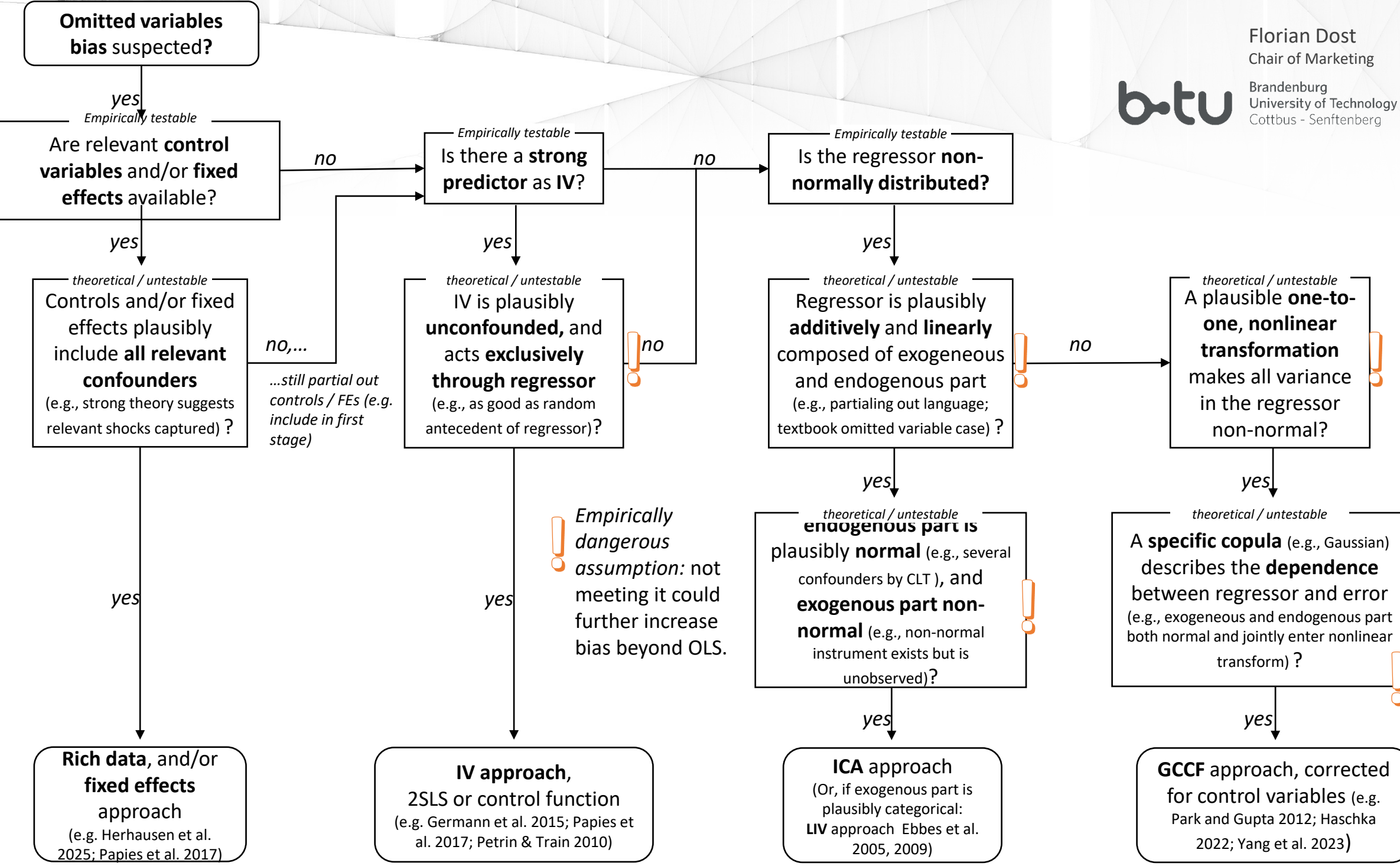
Authors

Steps:

- Define estimand, main threat, and confounders
- Ensure continuous endogenous regressor (ER) and dependent variable (DV), and sufficient (>250) obs
- Prevent/reduce bias before cure, then regress ER on fixed effects and exogenous covariates, save residuals
- Assess non-normality of (residual) ER: plot histogram, skewness, kurtosis; test against normal distribution (e.g. Kolmogorov-Smirnov and Jarque-Bera tests)
- Argue for additive DGP (which is the traditional textbook case) with normal confounder (e.g., using CLT), and non-normal exogenous variance (e.g., given if a non-normal IV or antecedent exists)
- Run two-components ICA on ER and DV using JADE, assess normality of both components, then save the (more) normal component (use K-S statistic) as control function term
- Run target model, including prevention (covariates, fixed effects) and control function term, bootstrap standard errors, report results and diagnostics

Hypothetical example as text template:

“We study the effect of review ratings (a continuous variable), as proxy for product quality $[P]$, on sales $[Y]$. A threat is omitted-variable bias arising from unobserved strategic behaviors such as managers pushing buyers for better ratings, and fake reviews (both harmful and beneficial), from manufacturer, seller, and possibly competitors [*confounders*]. Even with extensive controls and weekly fixed effects [*prevention*], residual confounding may remain. Unfortunately, we have no strong instrument available; yet the residualized review ratings display a non-normal distribution: (*skewness* = ..., *kurtosis* = ..., Jarque-Bera p = ..., Kolmogorov-Smirnov p = ...) [*non – normality*], allowing us to consider an instrument-free ICA method. Identification requires that the confounder(s) be approximately normal and, together with non-normal exogenous variation, additively determine the regressor. We assume reviewers' baseline product-quality assessments are independent, with managerial pressures entering additively. Both managerially affected and unaffected reviews are present. We also expect that our controls ‘partial out’ unobserved heterogeneity. All arguments motivate an additive DGP. Further assuming several confounding influences supports the idea of an overall approximately normal endogenous variance in the regressor. It is plausible, that the exogenous part reflecting perceived product quality is not normal for three reasons: effective regulation will affect the lower end, nonlinearly increasing cost the higher end of the quality distribution, leading to skew; reviewers perceptions and consequent ratings, may emphasize particularly negative or positive judgments, leading to excess kurtosis. We recover the normal component via JADE, then include the more normal component in the outcome equation as control function. Inference uses bootstrapped robust standard errors with 199 replications. Results are as follows...”



Reviewer perspective: what to ask?

Reviewers

- **Are estimand and main threats well defined and plausible?**
 - Consider coherence between estimand and known antecedents and consequences
 - Suggest obvious and plausible unmentioned confounders
- **Are available controls used, and used controls relevant?**
- **After prevention, does an omitted-variables threat plausibly remain?**
 - When convinced of unbiased estimation, suggest sensitivity or threshold test, without extra correction
- **Is the regressor plausibly and demonstratively non-normal?**
 - When in doubt, ask for multiple common normality test statistics and a histogram
 - Check if non-normality persists *after* partialling out relevant controls (they could correlate with regressor)
- **Is the DGP plausibly additive?**
 - This is the textbook default
 - Look for tell-tale signs such as use of controls, fixed effects, or PCA
- **Is the confounder plausibly normal?**
 - Consider known distributions in case of well-established confounders in extant studies
 - Several confounders in an additive DGP will always converge on a normal distribution due to the CLT
- **How is (normal) endogeneity introduced while the regressor became non-normal?**
 - Consider plausible reasons why the regressor exhibits skew or thin or fat tails from an exogenous antecedent
 - In case authors suggest use of GCCF, use extra scrutiny: GCCF require a DGP with a one-to-one, nonlinear transform; even with a stated transform, with any factor added after, or on top of the transform GCCF methods will not only fail to reduce bias, but likely lead to *worse* bias than OLS. Ask: is such a transform plausible and economically motivated?
- **Is the method well implemented with replicable results?**
 - ICA-based methods require robust standard errors from a bootstrap procedure
 - Check replication package and code
- **Was it worth the extra effort?**
 - Triangulate with other results (e.g., OLS, IV-based)
 - Every correction method is less efficient than OLS if not needed and should not be used "pro forma"

Not another IV-free endogeneity correction... Nevertheless, let's have a discussion!

Slide 27 Florian Dost
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The Gaussian copula method cannot
correct for omitted variables bias



Our ICA-based alternative to correct for
omitted-variable bias without instruments



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R function to apply our ICA-
based alternative is available
on GitHub!